

Rainbow Food List — AI-Based Diet Suggestion System

Project Requirements Document (PRD)

Field	Value
Document type	Product & Project Requirements (requirements-level only — no design/implementation)
Product name	Rainbow Food List
System name	AI-Based Diet Suggestion System
Version	1.0
Status	Draft for stakeholder review
Primary sources	Rainbow Food list Proposal Edited (.md/.pdf) , AI-Based Diet Suggestion System.docx
External research	Bangladesh Food Composition Table, chronic-disease dietary guidance, AI nutrition research, Bangladesh data-protection law (sources in Appendix B)

About this document. This PRD consolidates two source documents (a *Project Charter* and a *Proposal*) that describe the same idea under two names and with several conflicting figures. It restates the idea as a single, consistent set of **requirements** and a **conceptual description of how the system works**. It deliberately does **not** cover coding, technology design, or implementation detail. Conflicts found in the source material are flagged in **Section 14** and must be resolved by stakeholders.

1. Vision

Build a free-to-use, Bangladesh-first nutrition companion that gives any user accurate food-composition data and gives each user **personalized, safe diet guidance** based on their health profile, goals, and budget — with an optional paid tier for AI-generated diet charts and professional nutritionist support.

The long-term aim is to reduce diet-related harm from chronic disease (diabetes, hypertension/heart disease, kidney disease) by turning unfamiliar nutrition data into simple, affordable, culturally appropriate food choices.

2. Problem Statement & Context

- **NCD burden is high and rising in Bangladesh.** Non-communicable diseases — hypertension, diabetes, and chronic kidney disease — are a growing share of the national disease burden, with substantial undiagnosed prevalence, especially in rural areas. Diet is a primary modifiable risk factor.
- **Nutrition data is inaccessible to ordinary users.** Authoritative composition data (e.g., the *Food Composition Table for Bangladesh*) exists but is not presented in a consumer-friendly, searchable form, and rarely in Bangla with local portion sizes.
- **Generic diet advice is unsafe for chronic conditions.** A single "healthy diet" fails people with kidney disease (who must limit sodium, potassium, phosphorus, and protein by stage), diabetes (carbohydrate/glycemic control), or heart disease (sodium, saturated fat). Advice must be **condition-aware**.
- **Affordability is a real constraint.** Many users cannot afford animal-based protein or imported "health foods"; they need the **cheapest food that meets a nutritional need**, not just the "healthiest".
- **Existing apps are not localized.** Most trackers assume Western foods and portions; a few local apps are emerging, so **local food accuracy and condition safety** is the differentiator.

3. Goals & Objectives

ID	Objective
OBJ-1	Provide detailed, searchable nutritional information (energy, protein, carbohydrate, fat, fiber, vitamins, minerals) for local foods.
OBJ-2	Deliver personalized diet suggestions driven by user health data, goals, and dietary restrictions.
OBJ-3	Support chronic conditions (diabetes, heart disease, kidney disease) with condition-specific guidance and safety limits.
OBJ-4	Suggest low-cost, nutrient-dense and plant-based alternative foods that meet the same nutritional need.
OBJ-5	Track progress and give understandable insights through a dashboard.
OBJ-6	Offer a premium tier with AI-assisted diet charts and nutritionist support, sustaining the free tier.

4. Target Users & Personas

Persona	Description	Key needs
Health-conscious individual	Wants to improve diet, manage weight, eat better.	Food lookup, simple recommendations, progress tracking.
Chronic-condition patient	Diabetes, hypertension/heart disease, or kidney disease, at various stages.	Condition-safe food lists, nutrient limits, warnings.
Dietary-restricted / vegetarian	Avoids animal products or has allergies/intolerances.	Equivalent plant-based and safe alternatives.
Budget-conscious user	Low income; needs affordable nutrition.	Cheapest nutrient-dense options, price-aware suggestions.
Premium user	Wants a structured plan / human guidance.	AI diet chart, nutritionist review.
Content/Admin staff (internal)	Maintains food data and guidance.	Data management, review, publication workflow.

5. How the Idea Works (Conceptual Overview)

This is the **conceptual flow only** — no technology decisions are made here.

- User creates a profile.** Age, sex, height, weight, activity level, health goals, declared conditions (e.g., diabetes, CKD stage, heart disease), allergies, dietary restrictions, budget sensitivity, and language preference.
- User browses or asks for guidance.** They can look up any food's composition, or request a recommendation.
- The system computes the user's needs.** From the profile it derives targets and limits (e.g., daily energy and macronutrient targets; and for CKD, caps on sodium/potassium/phosphorus/protein).
- The recommendation engine matches foods to needs.**
 - It draws from a **curated food-composition database** (local foods, portions, and prices).
 - It applies **condition rules and safety constraints** so suggested foods do not violate a user's limits.
 - It ranks candidates by fit to goals, preference, availability, and **cost**.
 - It produces a **meal/diet suggestion** (foods, portions, and — in premium — a full diet chart).
- Alternative suggestions are offered.** If a food is unavailable, too expensive, or unsuitable, the system proposes equivalent alternatives (including plant-based) with similar nutrient contribution.
- User logs intake / follows the plan.** Logged consumption feeds the dashboard.

7. **The dashboard shows progress and insights.** Intake vs. targets, trends, warnings when a limit is approached, and plain-language feedback.
8. **Feedback loops back.** User feedback and outcomes inform periodic improvement of recommendations and content. Premium users may additionally have a nutritionist review their chart.

Trust & safety is built into the flow: every recommendation carries a *not-medical-advice* disclaimer, and condition-related suggestions are framed as guidance to discuss with a clinician.

6. Scope

6.1 In Scope (initial product)

- Curated food-composition database for commonly consumed local foods.
- User accounts, profiles, and health/dietary intake.
- Personalized diet recommendations (free tier: basic; premium tier: full AI diet chart).
- Condition-aware filtering/safety for diabetes, heart disease, and kidney disease.
- Plant-based / alternative and low-cost nutrient-dense suggestions.
- Progress dashboard and insights.
- Web + mobile access.
- Premium: AI-assisted diet charts and professional nutritionist review.
- Content/admin tooling to maintain the food database.

6.2 Out of Scope (initial product)

- Real-time human dietitian/video consultations.
- Fitness/workout tracking and exercise plans.
- Real-time integration with third-party health/wearable devices.
- Clinical diagnosis or treatment; the system does not replace medical care.
- Prescription or supplement dosing.

6.3 Roadmap (future, explicitly deferred)

- Wearable/fitness-device integration and holistic health insights.
- B2B/corporate wellness, white-label, and healthcare-provider partnerships.
- Deeper predictive analytics (e.g., long-term risk trends).

7. Functional Requirements

Priorities use MoSCoW: **M** = Must, **S** = Should, **C** = Could.

7.1 Accounts & Profiles

ID	Requirement	Priority
FR-1.1	Users can register, log in, and manage an account (email/phone/social login).	M
FR-1.2	Users can create/edit a health profile: age, sex, height, weight, activity level, goals (weight loss/gain/maintain, manage condition), declared conditions, allergies, dietary restrictions.	M
FR-1.3	Users can declare a budget sensitivity and preferred language (Bangla/English).	S

ID	Requirement	Priority
FR-1.4	Users can update their profile, and recommendations reflect changes.	M

7.2 Food & Nutrition Information

ID	Requirement	Priority
FR-2.1	Users can search/browse foods and view composition: energy, protein, carbohydrate, fat, fiber, vitamins, minerals.	M
FR-2.2	Data is presented per common local portion/household measure, not only per 100 g.	M
FR-2.3	Each food shows its key condition-relevant nutrients (sodium, potassium, phosphorus, saturated fat, added sugar, glycemic relevance).	M
FR-2.4	Foods indicate approximate local price/cost tier to support affordability guidance.	S
FR-2.5	Users can save/bookmark foods and view recently viewed items.	C

7.3 Personalized Recommendations

ID	Requirement	Priority
FR-3.1	System generates diet suggestions tailored to the user's profile, goals, and restrictions.	M
FR-3.2	Recommendations respect declared allergies/dietary restrictions (hard exclusions).	M
FR-3.3	Recommendations respect condition-specific limits for diabetes, heart disease, and kidney disease.	M
FR-3.4	System can generate a structured daily/weekly diet chart (premium).	M
FR-3.5	Suggestions include portion/quantity guidance.	S
FR-3.6	Users can regenerate/refine suggestions (e.g., swap a meal, exclude a food).	S
FR-3.7	System explains <i>why</i> a food was suggested or flagged (plain-language reason).	S

7.4 Alternatives & Affordability

ID	Requirement	Priority
FR-4.1	System suggests alternative foods with similar nutritional contribution when a food is unsuitable or unavailable.	M
FR-4.2	System suggests plant-based alternatives for users avoiding animal products.	M
FR-4.3	System highlights low-cost, nutrient-dense foods meeting a given need.	M
FR-4.4	Alternatives are ranked by cost and availability where budget sensitivity is set.	S

7.5 Progress Dashboard & Feedback

ID	Requirement	Priority
FR-5.1	Users can log meals/intake (search-based; optional barcode/photo in roadmap).	M
FR-5.2	Dashboard shows intake vs. targets (energy, macros, key condition nutrients) with charts.	M

ID	Requirement	Priority
FR-5.3	Dashboard shows trends over time and flags approaching/exceeded limits.	S
FR-5.4	Users can submit feedback and report a problem with a recommendation.	S
FR-5.5	Users can track adherence to their plan and view simple progress indicators.	C

7.6 Premium Services

ID	Requirement	Priority
FR-6.1	Premium users receive AI-assisted custom diet charts.	M
FR-6.2	Premium users can request professional nutritionist review of their chart (asynchronous).	M
FR-6.3	Subscription purchase/renewal and entitlements are managed in-app.	M
FR-6.4	Free vs. premium feature boundaries are clearly communicated.	S

7.7 Content & Administration

ID	Requirement	Priority
FR-7.1	Authorized staff can add/edit/retire food items and nutrient values.	M
FR-7.2	Data changes go through a review/approval step before publishing.	S
FR-7.3	Staff can manage condition guidance content and disclaimers.	M
FR-7.4	System tracks data provenance/version for each food record.	C

8. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	Recommendations return fast enough for interactive use (target: within a few seconds for a diet chart; near-instant for food lookups).
NFR-2	Scalability	Supports growth to a large public user base via elastic cloud infrastructure; food database and user data scale independently.
NFR-3	Availability	High uptime suitable for daily-use consumer app, with monitoring and alerting.
NFR-4	Security & Privacy	Encrypt sensitive health data in transit and at rest; strict access control; least-privilege admin access.
NFR-5	Compliance	Comply with health-data privacy obligations (Bangladesh Personal Data Protection Act, 2026; align with GDPR/HIPAA principles if serving those regions).
NFR-6	Usability & Accessibility	Simple, low-literacy-friendly UI; clear language; color contrast and text-size accessibility; works on low-end devices.
NFR-7	Localization	Full Bangla and English; local food names, portions, and units.
NFR-8	Safety	Every recommendation surfaces a <i>not-medical-advice</i> disclaimer; unsafe/edge combinations are blocked, not just warned.
NFR-9	Data Quality	Nutrient data traceable to credible sources; changes reviewed; known limitations (e.g., raw vs. cooked, seasonal variation) documented.

ID	Category	Requirement
NFR-10	Maintainability	Food data and condition guidance updatable without code changes; documented for handover.
NFR-11	Auditability	Recommendation history stored and inspectable by the user to support trust and support tickets.

9. Data Requirements

9.1 Data Sources

- **Primary:** *Food Composition Table for Bangladesh* (FCTB) and national nutrition resources (FAO/BIRDEM/INFS lineage) for local foods.
- **Secondary/supplementary:** international composition databases (e.g., USDA FoodData Central) for foods absent from FCTB.
- **Guidance content:** evidence-based condition dietary guidance (kidney: sodium/potassium/phosphorus/protein limits by CKD stage; diabetes: carbohydrate/glycemic control; heart: sodium and saturated fat).
- **Approximate local pricing** data for affordability features.

9.2 Data Quality Rules

- Each food record has a credible source citation and version.
- Handling of raw vs. cooked states and edible portions is explicit.
- Missing values are handled and marked, not silently zeroed.
- Nutrients relevant to conditions are always populated or explicitly marked as unknown.
- Data is validated by a domain expert (dietitian/nutritionist) before publication.

9.3 User Data Collected

Demographics, anthropometrics, health conditions, dietary preferences/allergies, logged intake, recommendation history, feedback, and subscription status. All classified as **sensitive** and governed by NFR-4/NFR-5.

10. Condition-Specific Domain Requirements

The system must encode, at minimum, the following guidance. Exact thresholds to be validated and versioned by qualified nutritionists.

Condition	Nutrients of concern	Required behavior
Diabetes	Carbohydrate load, added sugar, glycemic impact	Prefer lower-glycemic, higher-fiber options; distribute carbs; flag high-sugar items.
Heart disease / hypertension	Sodium, saturated fat	Prioritize low-sodium, low-saturated-fat foods; flag high-sodium items.
Chronic kidney disease	Sodium, potassium, phosphorus, protein	Apply stage-aware limits; flag high-potassium/phosphorus foods; adjust protein.
Allergies/intolerances	Condition-specific allergens	Hard exclusion — never suggest.

11. Service Tiers

- **Free tier:** nutrition lookup, basic recommendations, and the progress dashboard — free for all users.
- **Premium tier:** AI-assisted diet charts and professional nutritionist review.

12. Stakeholders & Team Roles

Roles required (from source docs): Project Manager/Product Owner; Backend Developer; Mobile App Developer (Android & iOS); Frontend Developer (web); ML/AI Engineer; Data Scientist; Data Engineer; UI/UX Designer; Food & Health Researcher; Nutritionist/Dietitian (domain expert); Database Administrator; DevOps Engineer; QA Engineer; Content Writer/Nutrition Specialist; Technical Writer; Marketing Manager; Customer Support; Financial Advisor/Accountant; Legal Advisor; Founders/Owners; Investors; Sponsors.

RACI-style responsibilities: the Project Manager owns delivery and coordination; the Nutritionist/Domain Expert owns the correctness/safety of dietary guidance; the ML/Data team owns recommendation quality; Legal owns privacy/compliance; Support/Operations own post-launch maintenance.

13. Phases & Timeline (consolidated from source)

13.1 Phases

Phase	Scope	Duration (source)
Phase 1	AI-powered recommendation engine (data prep, feature engineering, model build, experimentation/selection, deployment & testing)	196 days
Phase 2	User-friendly web/mobile application (design, front-end, back-end, AI integration)	151 days
Phase 3	Progress dashboard & insights (design, data collection/storage, front-end, testing/optimization)	137 days
Phase 4	Database for profiles/preferences/recommendations	~2–4 weeks per sub-task
Phase 5	User manual & documentation	~6–9 weeks total
Pilot	2-month pilot with selected users	Date in source: 05/05/2026

13.2 Delivery Targets (from source)

- Pilot engagement rate $\geq 70\%$.
- Recommendation accuracy $\geq 85\%$ against user health goals.
- Positive feedback on accuracy and usability during pilot.

14. Conflicts, Assumptions & Open Questions ⚠

The two source documents disagree on several points. These must be decided before the project proceeds.

#	Issue	Source A (Charter)	Source B (Proposal)	Needed decision
C-1	Project name	AI-Based Diet Suggestion System	Rainbow Food List	Confirm product name vs. system name (doc assumes both).
C-3	Timeline	Kick-off 15 Sep 2024; pilot criteria 2 months	Pilot date 05/05/2026	Confirm current schedule (both dates are now in the past).
C-4	Platform	"Web or mobile"	Mobile app + web interface	Confirm initial platform priority.
C-5	Human consultation	Out of scope (real-time dietitian)	Premium includes nutritionist consultation	Confirm async review in scope; real-time out.
C-6	Wearables	Out of scope	Listed as a revenue opportunity	Confirm wearables are roadmap-only.
C-7	Data sources	"Nutritional databases" (generic)	Not specified	Adopt FCTB as primary source (proposed here).
C-8	Accuracy target	85% recommendation accuracy	Not defined	Define how "accuracy" is measured and by whom.
C-9	Segment	Not defined	Subscription/freemium/ad-based	Confirm the free vs. premium feature split.
C-10	Region/regulatory	GDPR/HIPAA	Not stated	Decide launch region and governing law (Bangladesh PDPA 2026 proposed here).

Working assumptions in this PRD (until decided): product = Rainbow Food List; system = AI-Based Diet Suggestion System; launch region = Bangladesh; primary data = FCTB; free + premium freemium model; mobile-first with web; async nutritionist review in scope; wearables/diagnosis out of scope; not-medical-advice disclaimer always shown.

15. Risks & Mitigations

Risk	Impact	Mitigation
Inaccurate/unsafe recommendations	Harm to users, loss of trust	Curated reviewed data, condition rules, expert sign-off, disclaimers, feedback loop.
Poor data quality	Wrong nutrient values	Validation, provenance tracking, expert review, documented limitations.
Health-data breach / non-compliance	Legal and reputational damage	Encryption, access control, compliance with PDPA/GDPR/HIPAA principles, audits.
Team turnover	Knowledge loss	Documentation, cross-training, centralized knowledge base.
Low user adoption	No impact/adoption	Pilot with target users, iterate, differentiate on local accuracy + condition safety.
Regulatory / health-guideline non-compliance	Legal exposure	Legal review, adherence to published dietary guidance, versioned content.
Market competition	Overshadowed by competitors	Focus on local foods, affordability, and condition-aware safety as differentiators.
Over-reliance on AI	Wrong advice in edge cases	Human-in-the-loop for premium, expert rule constraints on top of model output.

16. Success Metrics

- **Pilot:** engagement $\geq 70\%$; recommendation accuracy $\geq 85\%$; positive usability/accuracy feedback.
- **Product:** active users, retention, premium conversion rate, recommendation acceptance/swap rate, data-coverage of common local foods.
- **Health impact (long-term):** improved adherence to plan; user-reported outcomes (with appropriate caveats about attribution).
- **Operations:** uptime, support response time, model refresh cadence.

17. Glossary

- **FCTB** — Food Composition Table for Bangladesh.
- **CKD** — Chronic Kidney Disease.
- **NCD** — Non-Communicable Disease.
- **PRD** — Product/Project Requirements Document.
- **PDPA** — Personal Data Protection Act (Bangladesh, 2026).
- **MoSCoW** — Must/Should/Could/Won't prioritization.

Appendix A — Requirement Traceability (source → this doc)

Source section	Mapped to
Charter §3 Scope / §4 Deliverables	Sections 6, 7
Charter §7 Risks	Section 15
Charter §8 Pilot	Sections 13, 16
Proposal §3 Features / §4 Audience	Sections 4, 7
Proposal §6 Hardware/Software	Section 8 (technical constraints only)

Appendix B — External Sources Consulted

- Intelligent diet recommendation system (AI/ML nutrition) — <https://www.sciencedirect.com/science/article/pii/S2405457725029249>
- AI nutrition recommendation using a deep generative model — <https://www.nature.com/articles/s41598-024-65438-x>
- AI-driven knowledge-based personalized nutrition (rule/constraint-based) — <https://www.frontiersin.org/journals/nutrition/articles/10.3389/fnut.2026.1894893/full>
- Food Composition Table for Bangladesh — https://www.fao.org/fileadmin/templates/food_composition/documents/FCT_10_2_14_final_version.pdf
- Diabetes & kidney disease: what to eat (CDC) — <https://www.cdc.gov/diabetes/healthy-eating/diabetes-and-kidney-disease-food.html>
- CKD nutrition: sodium, potassium, phosphorus, protein (National Kidney Foundation) — <https://www.kidney.org/kidney-topics/creating-kidney-friendly-plate>
- Non-communicable disease burden in Bangladesh (Lancet Global Health) — [https://www.thelancet.com/journals/langlo/article/PIIS2214-109X\(23\)00432-1/fulltext](https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(23)00432-1/fulltext)
- Bangladesh Personal Data Protection Act 2026 overview — <https://www.recordinglaw.com/world-laws/world-data-privacy-laws/bangladesh-data-privacy-laws/>
- Local competitor example (AI calorie tracker for Bangladeshi food) — <https://samikhanfitness.com/mepe-khai/>

End of Project Requirements Document.